

Bo Shang

Forward Deployed Engineer | Robotics Data & Applied AI
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Forward-deployed engineer who turns ambiguous field and stakeholder needs into working AI and robotics systems. Own the loop from requirements and data-collection specifications through Python pipelines, QA, cloud/edge deployment, and end-user delivery. Experienced with messy real-world sensor data, cross-configuration failures, and 0-to-1 products spanning robotics, infrastructure, and applied AI.

EXPERIENCE

- **University of Maryland — CEE / Maryland Transportation Institute** College Park, MD
Postdoctoral Associate Jul 2026 – Present
 - Own hardware/software integration, field-data support, testing, troubleshooting, and deliverables for two sponsored transportation projects; translate objectives and field constraints into executable test plans and acceptance evidence
- **ScholarMind AI / Merit Point Academy** Remote
Independent AI Product Engineer 2026 – Present
 - Designed and shipped a **live bilingual AI education and research platform** spanning an LLM research assistant, 34-course library, recorded lessons, short-video publishing, robotics evidence pages, and a LeetCode knowledge-graph game with 34 nodes / 102 progression units
 - Directed AI coding agents across requirements, architecture, implementation, testing, privacy/attribution guardrails, and production releases; integrated Claude and self-hosted Qwen/Ollama-backed workflows across multiple subdomains
- **City University of New York — AI & Mobility Research Lab** New York, NY
Field Data Engineer — Roadside Perception Jul 2025 – Jul 2026
 - Deployed an end-to-end **fixed LiDAR + camera pipeline at NYC DOT highway sites**; owned collection, temporal/spatial alignment, ingestion, and training-ready delivery
 - Converted field constraints into portable data/evaluation specifications; enabled models trained at one site to run across different beam counts, frame rates, and mounting geometries
 - Built multi-frame reconstruction and vulnerable-road-user labeling/QA workflows; turned recurring coverage, synchronization, and detection failures into reusable configuration, versioning, and conversion infrastructure
- **City University of New York — Robotics Lab** New York, NY
Field Data Engineer — Robotic Inspection Dec 2022 – Dec 2024
 - Owned end-to-end delivery of a **robotic bridge-inspection workflow**: robot/drone imagery and metadata, ingestion, model inference, 3D reconstruction, QA, and engineering-user review
 - Built a custom **WebODM platform** combining automated segmentation, interactive 3D visualization, defect localization, and crack measurement for city engineers and inspectors
 - Trained CNNs for crack, spalling, and stain detection; deployed inference to **AWS**, iterated against real-image failures, and coordinated robotics, structural-engineering, data, and software contributors
- **Missouri University of Science and Technology** Rolla, MO
Embedded Data Systems Engineer — Robotics Deployment Jan 2020 – Nov 2022
 - Deployed real-time perception on **NVIDIA Jetson** for an autonomous bridge-inspection drone; used VICON-grounded QA and field tests to isolate sensor, localization, control, power, latency, wind, and platform-motion failures

CORE COMPETENCIES

- **Customer & Delivery**: Discovery, requirements, data/annotation specifications, delivery sequencing, QA, stakeholder communication, troubleshooting, reusable playbooks
- **Robotics Data**: LiDAR, RGB, thermal, IMU; ROS / ROS 2, MCAP, Protobuf; synchronization, calibration, metadata, labeling, dataset versioning, cross-configuration validation
- **Engineering**: Python, C++, PyTorch, OpenCV, Linux; AWS (S3/EC2/SageMaker), Docker, CI/CD, NVIDIA Jetson, WebODM, model packaging and serving

EDUCATION

- **Northeastern University & University of California, Merced** Shenyang, China / Merced, CA
Ph.D. in Pattern Recognition & Intelligent Systems Aug 2013 – Dec 2020
- **Northeastern University** Shenyang, China
M.Eng. in Pattern Recognition & Intelligent Systems; B.Eng. in Automation Aug 2007 – Dec 2013

SELECTED EVIDENCE

- First/equal-first-author work in CVPR 2026 Workshops, IEEE T-ASE / IROS 2025, and MobiSPC 2025 on roadside LiDAR and robotic inspection
- **US 12,296,994** (granted May 2025) — Unmanned vehicle system design and deployment